

Cameron Mars

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Education

University of Colorado Boulder – College of Arts & Sciences

Graduating May 2027

Bachelor of Science in Mathematics

Minor in Computer Science

GPA: Overall 3.98 — Mathematics 4.0

Notable Coursework: Data Structures, Algorithms, Computer Systems, Software Development, Linear Algebra, Discrete Math, Differential Equations, Calculus Series, Analysis, Abstract Algebra, Cybersecurity, Principles of Programming Languages, Numerical Analysis

Work Experience

Calculus and Differential Equations Tutor – CU Boulder Athletic Department

June 2024 – Present

- Provide tutoring to college athletes across multiple levels of mathematics, focusing on Calculus, Differential Equations, and Linear Algebra
- Design mock exams and exam preparation courses
- Develop structured study plans tailored to course objectives

Student Researcher – CU Boulder Mathematics Department

May 2025 – April 2026

- Conduct original research in functional analysis on the classification of spatial partial isometries in finite-dimensional L^p spaces
- Proved that all partial isometries are spatial in this setting, extending classical operator theory results
- Developed new theorems including cardinality results for uniform partial isometries and generalizations of full isometry structure theorems

Algorithms Course Assistant – CU Boulder Computer Science Department

January 2026 – May 2026

- Course assistant for CSCI 3104: Algorithms
- Support student understanding of theoretical computation models, algorithmic complexity, dynamic programming, maximum flow algorithms, etc
- Assist with homework, study strategies, and exam preparation

Computer Systems Course Assistant – CU Boulder Computer Science Department

August 2025 – December 2025

- Course assistant for CSCI 2400: Computer Systems
- Support student understanding of binary operations, optimization, assembly, and machine-level programming
- Assist with homework, labs, study guides, and exam preparation

Algebra and Geometry Tutor – Manhattan School of Arts and Academics

January 2024 – May 2024

- Co-developed a Spanish-language tutoring program for middle school students
- Tutored disadvantaged grade 6–8 students in Algebra and Geometry
- Designed lesson plans focused on long-term academic success

Skills & Notable Projects

Linux File System Simulator

- Implemented a command-line file system simulator in C++ using an N-ary tree to represent directories and files
- Built a custom hash table with linear probing, quadratic probing, and chaining for fast file lookup
- Designed automated benchmarks to analyze collision rates, performance, and memory usage

SVD Decomposition Image Processor

- Compressed NOAA Mariana Trench depth data using incomplete SVD, reducing 1.9M points to 140K with < 0.5% error
- Applied eigenvalue decomposition and Gram–Schmidt orthogonalization
- Created MATLAB contour and surface plots to validate data fidelity

Revenge of Guts Game

- Developed a story-driven survival game in C++ with resource gathering, crafting, and boss combat
- Implemented object-oriented systems for inventory, training progression, and combat mechanics

Wheelchair-Accessible Camera Controller

- Engineered an Arduino-based joystick camera system for quadriplegic users
- Integrated mechanical and electrical components for smooth directional control
- Emphasized low-cost, inclusive, and user-friendly design

PowerPlay Website

- Built a web-based trading card game driven by real NBA/WNBA statistics
- Developed front-end UI and interactive battle logic in JavaScript
- Collaborated on full-stack features including authentication, trading, and game-state tracking

Technical: Scala, C/C++, HTML, CSS, JavaScript, SQL, Node.js, MATLAB, Arduino, Microsoft Office

Languages: Spanish, German

Leadership & Volunteer Experience

Theta Tau Engineering Fraternity

August 2024 – Present

- Support planning of service, professional development, and brotherhood events
- Mentor new members and provide peer tutoring in engineering coursework

CU Boulder Arts and Sciences Student Government

August 2023 – May 2024

- Assisted with allocation of student organization funding
- Participated in planning campus engagement and well-being initiatives
- Collaborated with administrators to address student concerns

CU Boulder Alternative Breaks

March 2024

- Traveled to New Mexico to support underprivileged immigrants
- Assisted with food preparation, facility maintenance, and technology education

Hobbies & Interests

Basketball, Muay Thai, Running, Skiing, Weightlifting, Philosophy, Classic Literature, Anime/Manga